

Exploratory Data Analysis (EDA) Summary Report

Geldium Finance: Credit Card Delinquency Risk | Prepared by Tata iQ

1. Introduction

This report presents the exploratory data analysis (EDA) of Geldium Finance's customer dataset, carried out by Tata iQ before any delinquency prediction model is built. The goals are to (1) assess the structure, completeness and quality of the data, (2) decide how missing and inconsistent values should be treated, and (3) identify early risk indicators and anomalies that the Analytics team and Geldium's Collections leadership should consider when designing the predictive model and intervention strategies.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key dataset attributes:

- **Number of records:** 500 customers (one row per customer, 19 columns). No duplicate rows and no duplicate Customer_IDs.

- **Key variables:** Age, Income (annual, USD), Credit_Score (300-850), Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Employment_Status, Account_Tenure (years), Credit_Card_Type, Location, Month_1 to Month_6 (six-month payment history) and the target Delinquent_Account (0/1).

- **Data types:** 8 numerical (Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure), 1 binary target (Delinquent_Account), 4 categorical (Customer_ID, Employment_Status, Credit_Card_Type, Location) and 6 ordinal payment-history fields (Month_1 to Month_6: On-time / Late / Missed).

- **Target balance:** 80 of 500 customers (16%) are delinquent and 420 (84%) are not, so the classes are imbalanced.

Data quality summary:

The dataset is structurally clean (500 unique customers, no duplicate rows, and no impossible ages, credit scores or income values) but has several issues to resolve before modelling. Income (7.8%), Loan_Balance (5.8%) and Credit_Score (0.4%)

contain missing values, and Employment_Status uses six inconsistent labels for four categories. Utilization and debt-to-income are stored as fractions instead of the documented percentages, four accounts exceed 100% utilization, and Missed_Payments matches the monthly payment history in only 14% of records. Delinquent accounts make up just 16% of the data, so the model will need imbalance-aware training and evaluation.

Key inconsistencies and anomalies:

Issue	What was observed	Proposed action
Employment labels	Six raw labels (Employed, employed, EMP, Self-employed, Unemployed, retired) for four groups; 'Retired' is not in the data guide examples.	Standardize to Employed / Self-Employed / Unemployed / Retired.
Units	Credit_Utilization (0.05-1.03) and Debt_to_Income_Ratio (0.10-0.55) are fractions, but the guide says 0-100%.	Rescale to % or update the data dictionary.
Utilization above 100%	4 customers exceed 100% (max 102.6%).	Confirm if over-limit or error; keep with a flag.
Missed_Payments vs history	Equals the count of 'Missed' months in Month_1-6 for only 14.4% of customers ($r = 0.01$). Guide says past 12 months but the maximum is 6.	Clarify definition and window with Geldium.
Payment history coding	Month_1-6 are text (On-time / Late / Missed), not 0/1/2 as in the guide.	Encode as ordinal 0/1/2.
Income vs employment	84 of 93 'Unemployed' customers report income; median income is similar across all employment groups (about \$102k-\$109k).	Clarify what Income captures (household, pensions, benefits).
Tenure vs age	81 customers have Account_Tenure greater than (Age - 18), e.g. tenure longer than adult life.	Validate source fields; flag for review.
Score distribution	Credit_Score is spread almost evenly from 301 to 847 (about 30% below 450), unlike typical lending portfolios.	Check whether data is simulated or sampled.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- **Variables with missing values:** Income (39 rows, 7.8%), Loan_Balance (29 rows, 5.8%) and Credit_Score (2 rows, 0.4%). In total 64 of 500 rows (12.8%) have at least one missing value; all other columns are complete.

- **Missing data treatment:** Median imputation for all three variables, with missing-value indicator flags for Income and Loan_Balance. No columns are dropped and no synthetic data is needed.

Missing data handling table:

Missing data issue	Handling method	Justification
Income (39 missing, 7.8%)	Median imputation (\$107,658) plus missing flag	Distribution is near symmetric (skew about 0.05); missingness is unrelated to employment status ($p = 0.44$) or delinquency ($p = 0.82$), consistent with MCAR; under 8% missing.
Loan_Balance (29 missing, 5.8%)	Median imputation (\$45,776) plus missing flag	Missing rows show higher delinquency (24.1% vs 15.5%, not significant at $p = 0.29$ with a small sample), so the flag lets the model detect any MAR/MNAR effect.
Credit_Score (2 missing, 0.4%)	Median imputation (586)	Only 2 records, so the effect is negligible and imputation keeps the rows.

Synthetic data was considered but not recommended for these gaps: they are small (under 8%) and transparent statistical imputation is more explainable and regulator-friendly. Synthetic records could later be used to stress-test the model or to help with the 16% delinquent class, but only if validated against real distributions. Imputation values should be calculated on the training split only to avoid data leakage.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

- **Correlations observed between key variables:** Linear correlations between Delinquent_Account and every numeric variable are very weak ($|r|$ of 0.05 or less, none statistically significant, all $p > 0.3$). Credit_Utilization (0.03), Debt_to_Income_Ratio (0.03), Credit_Score (0.03) and Income (0.05) lean slightly positive, while Missed_Payments (-0.03) and Account_Tenure (-0.04) lean slightly negative. Several of these run opposite to business intuition.

- **Segment-level differences (indicative only, small samples):** Unemployed customers 19.4% delinquent vs Retired 11.5%; Business cards 21.3% vs Platinum

11.8%; Los Angeles 19.6% vs New York 12.0%; utilization up to 30% at 12.0% vs 16-17% in higher bands; ages 45-60 at 21.7% vs 12.3% for ages 31-45.

- **Unexpected anomalies:** Delinquent customers have slightly higher average credit scores (591 vs 575) and income (\$113.9k vs \$107.3k) than non-delinquent ones; customers with 0 missed payments show 18.2% delinquency vs 11.5% for those with 6; and the payment-history months split roughly one third On-time / Late / Missed in every month with no worsening trend (delinquent accounts average 3.96 late or missed months vs 4.04 for others). Together with the even credit score spread, this suggests the data may be simulated, or that Delinquent_Account is defined differently from the payment fields. This needs confirmation from Geldium before modelling.

High-risk indicators for delinquency prediction:

- **Payment history (Month_1-6, Missed_Payments):** Past late or missed payments are normally the strongest predictor of future default, but they show no relationship here, so data reliability must be verified first.

- **Credit utilization:** High utilization signals financial stress; delinquency rises from 12.0% (up to 30% utilization) to 16-17% in higher bands.

- **Debt-to-income ratio:** A high ratio limits repayment capacity; it is slightly higher for delinquent customers (30.6% vs 29.7%).

- **Employment status:** Unemployed customers show the highest delinquency (19.4%), supporting income stability as a risk factor.

- **Card type and account tenure:** Business (21.3%) and Student (17.9%) cards and short tenure (17.8% for up to 3 years vs 14.0% beyond 13 years) point to higher risk in newer or less established accounts.

- **Credit score:** Normally a core risk measure, but it currently shows no expected relationship and needs investigation.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

Example AI prompts used:

- 'Summarize key patterns in the dataset and identify anomalies.'

- 'Suggest an imputation strategy for missing income values based on industry best practices.'

- 'Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.'
- 'Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning.'
- 'Check whether missing Income and Loan_Balance values are related to employment status or delinquency (MCAR, MAR or MNAR).'
- 'Compare delinquency rates across utilization, employment status, card type and age bands, and flag results that contradict expected risk logic.'

Only the anonymised dataset (pseudonymous Customer_IDs, no names or contact details) was analysed. AI-generated figures were verified with direct calculations and statistical tests, and imputation decisions are to be validated by analysts and domain experts rather than left to GenAI. No synthetic data was used in this analysis.

6. Conclusion & Next Steps

After light cleaning, the dataset is usable: 87% of records are fully complete and missing values are limited to three fields that can be handled with transparent median imputation. The main concerns are data consistency (labels, units, and the mismatch between Missed_Payments and monthly history) and the very weak observable link between the available variables and delinquency, which suggests the data definitions should be validated before model building.

Recommended next steps:

- Standardize Employment_Status, convert utilization and DTI to percentages, and encode Month_1-6 as ordinal values.
- Confirm with Geldium the definitions of Missed_Payments (12 vs 6 months) and Delinquent_Account, and clarify Income for unemployed customers and tenure vs age.
- Apply median imputation with missing flags, fitted on the training split only.
- Build interpretable baselines (logistic regression, decision tree) with stratified splits and imbalance handling, evaluated with AUC-ROC, precision and recall.
- Run fairness checks (for example disparate impact across age and location) and avoid demographic variables driving decisions directly.
- Use synthetic data only for stress-testing, validated against real distributions.